

SALAMANDERS AND NEWTS

Unlike their fellow amphibians, the frogs, salamanders and newts normally have a slender, lizard-like body, a long tail, and four legs similar in size.

Newts and salamanders, also known as urodeles, are generally found in damp habitats and are largely confined to the northern hemisphere. They are numerous in the Americas, ranging from Canada to northern South America. They vary considerably in size, from species over 1 m (3¼ ft) in length, to tiny creatures about 2 cm (¾ in) long.

AMPHIBIOUS LIFESTYLES

Some species, notably the newts, spend part of their life in water, part on land. Some salamander species live their entire lives in water, while others are wholly terrestrial. Most have smooth, moist skin through which they breathe to a greater or lesser extent.

The salamanders of one family, the plethodontids, have no lungs and breathe entirely through their skin and the roof of their mouth. Urodeles have a relatively small head, compared with frogs and toads; they also have smaller eyes, smell being

the most important sense used in finding food and in social interactions. Most species, especially the terrestrial ones, are nocturnal, hiding under a log or rock during the day.

REPRODUCTION

In the majority of species, the eggs are fertilized inside the female. Males, however, do not have a penis but package sperm in capsules called spermatophores which are passed to the female during mating. In many species sperm transfer is preceded by elaborate courtship in which the male induces the female to co-operate with him. She may, of course, reject his advances. In many newts, males develop dorsal crests and bright colours in the breeding season.

Many species lay their eggs in water. When the larvae hatch, they have a long, slender body, a deep, fin-like tail, and large, feathery external gills. The larvae are carnivorous, feeding on tiny water creatures. The exceptions to this rule are the wholly terrestrial species of salamander. These lay their eggs on land and the larval stage is completed within the egg, which hatches to produce a miniature adult.

PHYLUM	CHORDATA
CLASS	AMPHIBIA
ORDER	CAUDATA
FAMILIES	10
SPECIES	754



A male alpine newt sniffs a female before courting her. Odour helps to identify the gender and species of potential partners.

SIRENS

The wholly aquatic salamanders of the family Sirenidae are found in the southern USA and Mexico. They retain larval features in the adult stage and resemble eels.

They have external gills and no hind limbs.

smooth, slimy skin

50–90 cm
20–35 in

GREATER SIREN

Siren lacertina

This species with tiny forelimbs and a fin-like tail inhabits shallow rivers, lakes, and ponds in the S.E. USA and N.E. Mexico.

NEWTS AND EUROPEAN SALAMANDERS

The Salamandridae are small- and medium-sized salamanders and newts found in Europe, North Africa, Asia, and North America. Eggs are fertilized inside the female, with sperm being transferred from the male in a spermatophore (sperm capsule) during an elaborate courtship.

12–20 cm
4¾–8 in

CALIFORNIA NEWT

Taricha torosa

This nocturnal newt enters ponds in spring to mate and lay eggs. It secretes a lethal nerve toxin to discourage predators.

TORRENT SALAMANDERS

Confined to the extreme northwestern USA, the Rhyacotritonidae are a small family of four species. These stocky, semi-aquatic salamanders lay their eggs on rocks under water and have aquatic larvae.

COLUMBIA TORRENT SALAMANDER

Rhyacotriton kezeri

Confined to the forests of Oregon and Washington, this salamander lays its eggs in springs. Intensive logging has caused its numbers to decline considerably.

7.5–11.5 cm
3–4½ in

orange poison glands

12–18 cm
4¾–7 in

CROCODILE NEWT

Tylotriton verrucosus

Found in C. Asia, this robust newt breeds in ponds after the monsoon. Its orange markings signal distasteful secretions.

broad head with rounded jaws

flattened tail

15–30 cm
6–12 in

SHARP-RIBBED SALAMANDER

Pleurodeles waltl

This large Spanish and Moroccan species has a unique mode of defence. When grasped, it pushes out the sharp tips of its ribs through its skin.